CLASS 10 MATHEMATICS • CHAPTER 9

Some Applications of Trigonometry

Exhaustive point-by-point presentation covering heights, distances, line of sight, angle of elevation, angle of depression, and solved examples directly from the textbook.

🎓 Akash Srivastva (www.akashmaths.online)

Line of Sight & Angles

Line of Sight

- The line drawn from the eye of an observer to the point in the object viewed.
- Forms the hypotenuse or side of the imagined right-angled triangle in real-world scenarios.

Angle of Elevation vs Depression

- **Elevation:** Formed when looking *above* the horizontal level (raising head).
- **Depression:** Formed when looking *below* the horizontal level (lowering head).

Section 9.1 Live Check

Q1. How is the line of sight defined when an observer views an object?

Q2. What is the key difference between the angle of elevation and the angle of depression?

Q3. When looking down from a balcony to a flower pot, which angle is formed with the horizontal level?

Worked Exercise Highlights

Problem 1: Tower & Ground Point (Example 1)

- Distance from foot = 15m, angle of elevation = 60°.
- tan 60° = (Height AB) / (Base BC).
- √3 = AB / 15.
- **Height AB = 15√3 m.**

Problem 2: Electrician Pole (Example 2)

- Total height = 5m, reach point is 1.3m below top ⇒ BD = 3.7m.
- Angle with horizontal = 60°.
- sin 60° = BD / Ladder (BC) ⇒ 3.7 / BC = √3 / 2.
- **Ladder length BC ≈ 4.28 m.**

Exercise Highlights Live Check

Q1. Which trigonometric ratio is most helpful when you know the adjacent side and want to find the opposite side?

Q2. In ladder problems, why do we subtract the distance from the top when the worker stops below the summit?

Q3. Why is drawing a clear right triangle sketch essential before attempting height and distance problems?

Chapter Recap & Checklist

Basic Terminology

- **Line of Sight:** Line from eye to observed point.
- **Angle of Elevation:** Looking upward above horizontal level.
- **Angle of Depression:** Looking downward below horizontal level.

Problem Solving

- **Techniques:** Use right triangle trigonometric ratios (sin, cos, tan) to compute unknown heights and distances.

THANK YOU

"Trigonometry is a majestic structure built from the simple geometry of the right triangle." — Anonymous

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